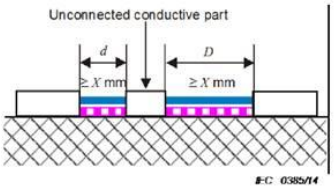
CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	Subclause(s)	Tracking No.	Publication date
IEC 62368-1:2014 ed.2.0 IEC 62368-1:2018 ed.3.0 IEC 60664-1:2020	ANNEX O	DSH 2214	2023
Category			
ITAV			
Subject	**Keywords**		**Developed by**
Measurement of creepage distances and clearances	Creepage, clearance		ETF2

Question

In IEC 62368-1:2014 ed.2.0 Annex O



Condition: Insulation distance with intervening, unconnected conductive part.

Rule: **Clearance** is the distance $d + D$, **creepage distance** is also $d + D$. Where the value of d or D is smaller than X mm it shall be considered as zero.

Figure O.4 – Intervening unconnected conductive part

In the figure above, there is an explanation indicates that:

Rule: Clearance is the distance d + D, creepage distance is also d + D. Where the value of d or D is smaller than X mm it shall be considered as zero.

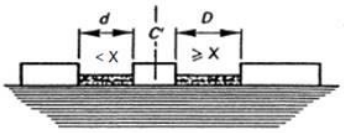
In IEC 62368-1:2018 ed.3.0 Annex O, there is also same expression.

Although, In IEC 60664-1: 2007 and IEC 60664-1:2020, there is no expression in this version of IEC 60664-1:2007 about the situation when d or D<X.

But, DSH 2160 clarifies the dilemma as below and you can see the decision sheet screen shot:

Disclaimer: This document is controlled and has been released electronically.
Only the version on the IECEE Website is the current document version

A PCB sample is tested according to sub-clause 6.2 of IEC 60664-1:2007. How to measure creepage distances when the path is split by floating conductive parts when d < X and D ≥ X?



NOTE: d < X, D ≥ X, C is conductive floating part
Figure test PCB sample

Different interpretations of total creepage distance are as follows:

Opinion 1: The creepage distance is measured as shown in IEC 60664-1:2007 example 11. Creepage distance is the distance = d + D.

Opinion 2: Since the d is less than X, the d is considered as zero. Creepage distance is the distance = D.

Which opinion is correct?

Decision

Opinion 1 is correct. Creepage distance is the distance = d + D.

Question is, how shall we evaluate clearance and creepage distance according to IEC 62368-1 for unconnected conductive parts (floating conductors)?

Decision

DSH 2160 is to be considered.

Explanatory notes

Figure O.4 is changed in 62368-1:2023 and is in line with DSH 2160 for 60664-1.

IEC TC108 HBSDT San Francisco Nov 2022: Agreement on the OSM EE document 22/5 with the same content.

Disclaimer: This document is controlled and has been released electronically.
Only the version on the IECEE Website is the current document version